

# **A Level Computer Science (9618)**

Topical Past Paper Worksheet

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## **Topic 3: Hardware**

Subtopic 3.1: Computers and their Components

### **Concept 3.1.4: Buffers**

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Total Questions: 7

**Mark Schemes begin on Page 6**

**Question 1 | Source: 9618\_w25\_qp\_11 (Q10.b)**

10 An architect needs a model of a building.

(b) State why a 3D printer needs a buffer.

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..... [1]

**Question 2 | Source: 9618\_w24\_qp\_11 (Q3.d.ii)**

3 A computer system has a dual-core Central Processing Unit (CPU).

(d) The computer system is used to store data received from a temperature sensor every five seconds. The data is stored on an optical disc using an optical disc reader/writer.

(ii) The computer uses a buffer when writing data to the optical disc.

Explain the use of a buffer when writing data to the optical disc.

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..... [3]

**Question 3 | Source: 9618\_w24\_qp\_13 (Q2.b)**

2 A memory buffer uses Dynamic RAM (DRAM).

(b) Explain how a memory buffer is used when a computer is transferring data to its magnetic hard disk drive.

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..... [4]

**Question 4 | Source: 9618\_s24\_qp\_11 (Q2.c.iii)**

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:
- The digital camera in the doorbell starts recording a video.
  - A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell’s internal secondary storage device and overwritten when the secondary storage device is full.



(c) The video doorbell has both primary memory and secondary storage.

(iii) The video doorbell uses a buffer.

Describe how the video doorbell will use the buffer.

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..... [2]

**Question 5 | Source: 9618\_s24\_qp\_12 (Q2.b)**

- 2 A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

- (b) The computer uses a buffer when transmitting data to the VR headset.

Explain how a buffer is used when data is transmitted between the computer and the VR headset.

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**Question 6 | Source: 9618\_w23\_qp\_13 (Q7.c)**

- 7 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (c) Describe how the laser printer makes use of a buffer.

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**Question 7 | Source: 9618\_w22\_qp\_13 (Q1.c)**

- 1 A digital audio message needs to be recorded.

(c) A computer uses a buffer when playing the audio message.

Explain the purpose of a buffer in a computer system using **one other** example.

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..... [3]

# MARK SCHEMES

## Mark Scheme for Question 1 | Source: 9618\_w25\_ms\_11 (Q10.b)

|       |                                                                                                                                                                                                                                                                                                                      |          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(b) | <b>1 mark</b> <ul style="list-style-type: none"> <li>To free up the processor to carry out other tasks as the rate data is received by the printer is different from the rate at which it can be processed</li> <li>To manage the mismatch in speed between the processor and the peripheral / 3D printer</li> </ul> | <b>1</b> |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

## Mark Scheme for Question 2 | Source: 9618\_w24\_ms\_11 (Q3.d.ii)

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(d)(ii) | <b>1 mark for each bullet point (max 3)</b> <ul style="list-style-type: none"> <li>The computer and optical disc reader/writer send and receive at different speeds</li> <li>A buffer allows temporary storage of the data</li> <li>... so that the computer can transfer data to the buffer at the higher speed</li> <li>... and is not held up waiting for data to transfer // so the computer can carry on with other tasks</li> <li>... so the optical disc reader/writer is not overloaded</li> <li>... and so that data is transferred to optical disc reader/writer from the buffer at the slower rate</li> </ul> | <b>3</b> |
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## Mark Scheme for Question 3 | Source: 9618\_w24\_ms\_13 (Q2.b)

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| 2(b) | <b>1 mark for each bullet point (max 4)</b> <ul style="list-style-type: none"> <li>The computer and the hard disk drive transmit and receive at different speeds // The computer transfers data faster than the HDD can receive</li> <li>The buffer is used for temporary storage</li> <li>... so that the computer can transfer data to the buffer at the higher speed</li> <li>... and is not held up waiting for data to transfer</li> <li>... and so that data is transferred to hard disk drive from the buffer at the slower rate</li> </ul> | <b>4</b> |
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### Mark Scheme for Question 4 | Source: 9618\_s24\_ms\_11 (Q2.c.iii)

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| 2(c)(iii) | <b>1 mark each to max 2:</b> <ul style="list-style-type: none"> <li>• Captured video is transmitted to buffer</li> <li>• ... video is transmitted from buffer to smartphone</li> <li>• Store recent data in a buffer for the user to rewind</li> <li>• ... instead of storing everything in secondary storage</li> <li>• Store readings from motion sensor</li> <li>• ... until the microprocessor can process them</li> <li>• Store video from digital camera</li> <li>• ... before moving it to secondary storage</li> </ul> | <b>2</b> |
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### Mark Scheme for Question 5 | Source: 9618\_s24\_ms\_12 (Q2.b)

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
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| 2(b) | <b>1 mark each to max 3:</b> <ul style="list-style-type: none"> <li>• The buffer is used as a <b>temporary</b> store for data going to the headset</li> <li>• Data is <b>transferred</b> into the buffer by the computer</li> <li>• Data is <b>retrieved</b> from the buffer by the headset</li> <li>• When the buffer is empty/full an interrupt is sent to the computer requesting more data/stopping further data being sent</li> <li>• When the headset has enough data/needs more data, an interrupt is sent by the headset to the computer to stop sending data from buffer</li> </ul> | <b>3</b> |
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### Mark Scheme for Question 6 | Source: 9618\_w23\_ms\_13 (Q7.c)

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| 7(c) | <b>1 mark for each bullet point (max 4)</b> <ul style="list-style-type: none"> <li>• The print instructions and data are sent by the laptop to a buffer (at laptop speed)</li> <li>• The data is transferred from the buffer to the printer (at printer speed)</li> <li>• ...allowing user to continue using the laptop // and allowing processor to continue processing</li> <li>• ...instead of waiting for relatively slower printer</li> <li>• When the buffer is empty an interrupt is sent to the laptop</li> <li>• ...requesting more data</li> </ul> | <b>4</b> |
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**Mark Scheme for Question 7 | Source: 9618\_w22\_ms\_13 (Q1.c)**

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| 1(c) | <p><b>1 mark</b> for purpose (<b>max 2</b>):</p> <p>Purpose:</p> <ul style="list-style-type: none"><li>• to act as temporary storage // to store (downloaded) data</li><li>• ... before it is used by the receiving device</li><li>• ... to allow processes / devices to operate at different speeds // independently of each other</li></ul> <p><b>1 mark</b> for each example (<b>max 1</b>):</p> <p>Examples:</p> <ul style="list-style-type: none"><li>• printer buffer used when data is transferred from a computer to a printer</li><li>• video buffer when streaming videos</li><li>• keyboard buffer when performing data entry</li></ul> | <b>3</b> |
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